HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - IV

Binomial, Poisson and Normal Distributions

Topics covered in this unit :

- Probability distributions: Binomial distribution
- Poisson distribution; Poisson approximation to the binomial
- Normal distribution and its properties
- Mean, variance, M.G.F., applications and numerical problems

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Probability Distribution - Overview

A probability distribution gives the probabilities of all values of a random variable. Discrete: Binomial, Poisson (p.m.f.). Continuous: Normal, uniform, exponential (p.d.f.).

Distribution	Type	Parameters	Mean	Variance
Binomial B(n, p)	Discrete	n, p	n p	n p q
Poisson P(lambda)	Discrete	lambda	lambda	lambda
Normal N(mu, sigma^2)	Continuous	mu, sigma	mu	sigma^2
Uniform U(a, b)	Continuous	a, b	(a+b)/2	(b-a)^2/12
Exponential	Continuous	theta	1/theta	1/theta^2

2. Binomial Distribution

2.1 Bernoulli trials

A trial with only two outcomes - success (probability p) and failure (probability q = 1 - p). Repeated n independent trials with the same p -> binomial setting.

Conditions for binomial distribution

- Fixed number n of trials
- Each trial has two outcomes: success / failure
- Trials are independent
- Probability of success p is constant

2.2 p.m.f.

$$P(X = x) = C(n, x)\, p^x\, q^{(n - x)}, \quad x = 0, 1, 2, \ldots, n$$

Notation: $X \sim B(n, p)$. Sum of probabilities $= (q + p)^n = 1$. (Binomial

expansion of (q + p)^n, hence the name.)

2.3 Constants

Quantity	Value
Mean	mu = n p
Variance	sigma^2 = n p q (variance < mean)
Standard deviation	sqrt(n p q)
Mode	integer part of (n + 1) p
Recurrence	P(x + 1) = [(n - x)/(x + 1)] (p/q) P(x)
M.G.F.	M(t) = (q + p e^t)^n
Skewness / Kurtosis	(q - p)/sqrt(npq) ; 3 + (1 - 6pq)/(npq)

2.4 Derivation of mean and variance

E(X) = Sum x C(n,x) p^x q^(n-x) = n p Sum C(n-1, x-1) p^(x-1) q^(n-x) = n p (q + p)^(n-1) = n p . E[X(X-1)] = n(n-1)p^2 -> Var = n(n-1)p^2 + np - n^2p^2 = n p q .

Example 1

A fair coin is tossed 6 times. Find P(exactly 4 heads), P(at least 4 heads).

n = 6, p = 1/2: P(4) = C(6,4)(1/2)^6 = 15/64 = 0.2344 . P(X >= 4) = (15 + 6 + 1)/64 = 22/64 = 0.34375 .

Example 2

The probability that a bulb is defective is 0.1. In a sample of 10, P(no defective) = (0.9)^10 = 0.3487 ; P(at most 2 defective) = 0.3487 + 10(0.1)(0.9)^9 + 45(0.01)(0.9)^8 = 0.3487 + 0.3874 + 0.1937 = 0.9298 .

Example 3 - mean and variance given

Mean of a binomial = 4 and variance = 3. Then n p = 4, n p q = 3 -> q = 3/4, p = 1/4, n = 16. P(X = 2) = C(16,2)(1/4)^2(3/4)^14 = 120 x 0.0625 x 0.01782 = 0.1336 .

Example 4 – expected frequencies

Out of 800 families with 5 children each, how many are expected to have 3 boys? $p = 1/2$: $P(3) = C(5,3)/32 = 10/32$. Frequency $= 800 \times 10/32 = 250$ families.

Example 5

A multiple-choice test has 5 questions, each with 4 options. If answered by guessing, P(at least 4 correct) $= C(5,4)(1/4)^4(3/4) + (1/4)^5 = 15/1024 + 1/1024 = 16/1024 = 0.0156$.

3. Poisson Distribution

Introduced by S. D. Poisson (1837). Used for rare events in a fixed interval of time / space: number of accidents per day, telephone calls per minute, misprints per page, defects per metre of cloth, radioactive decays.

3.1 p.m.f.

$$P(X = x) = e^{-\lambda}\lambda^x / x!, \quad x = 0, 1, 2, \ldots$$
$$(\lambda > 0)$$

Notation: $X \sim P(\lambda)$, where λ = average number of occurrences.

Quantity	Value
Mean	λ
Variance	λ (mean = variance)
Standard deviation	$\sqrt{\lambda}$
Mode	integer part of λ (if λ is an integer: λ and $\lambda - 1$)
Recurrence	$P(x+1) = [\lambda/(x+1)]\,P(x)$
M.G.F.	$M(t) = \exp[\lambda(e^t - 1)]$
Additive	$X_1 \sim P(\lambda_1)$, $X_2 \sim P(\lambda_2)$ independent $\Rightarrow X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$

3.2 Poisson as a limit of the binomial

Poisson approximation to the Binomial

- If n is large (n -> infinity), p is small (p -> 0), and n p = lambda stays finite, then
- C(n,x) p^x q^(n-x) -> e^(-lambda) lambda^x / x!
- Rule of thumb: n >= 20 and p <= 0.05 (or n p < 5)
- Proof idea: q^(n-x) = (1 - lambda/n)^(n-x) -> e^(-lambda), and n(n-1)...(n-x+1)/n^x -> 1

Derivation (short)

P(x) = [n(n-1)...(n-x+1)/x!] (lambda/n)^x (1 - lambda/n)^(n-x) = (lambda^x/x!) [1(1 - 1/n)...(1 - (x-1)/n)] (1 - lambda/n)^n (1 - lambda/n)^(-x). As n -> inf: product -> 1, (1 - lambda/n)^n -> e^(-lambda), last factor -> 1. Hence P(x) -> e^(-lambda) lambda^x / x! .

Example 1

A shop receives on average 3 calls per hour. P(exactly 2 calls in an hour) = e^-3 3^2/2! = 0.0498 x 4.5 = 0.2240 . P(no call) = e^-3 = 0.0498 . P(at least 1) = 0.9502 .

Example 2 - approximation

1 % of items are defective, in a sample of 200 find P(exactly 3 defective). n = 200, p = 0.01, lambda = 2. P(3) = e^-2 2^3/3! = 0.1353 x 8/6 = 0.1804 .

Example 3

A book of 500 pages has 500 misprints distributed randomly. Average = 1 per page. P(page has at least 2 misprints) = 1 - P(0) - P(1) = 1 - e^-1(1 + 1) = 1 - 0.7358 = 0.2642 . In a book of 1000 such pages: 264 pages expected.

Example 4 - mean and variance

If X ~ Poisson and P(X = 1) = P(X = 2): lambda e^-l = l^2 e^-l/2 -> lambda = 2 . Then P(X = 4) = e^-2 2^4/4! = 0.1353 x 16/24 = 0.0902 ; mean = variance = 2.

Example 5 - cars

The number of accidents on a road per week is Poisson with mean 2. P(at most 2 accidents in a week) = e^-2 (1 + 2 + 2) = 0.1353 x 5 = 0.6767 .

Basis	Binomial	Poisson
Trials	Fixed n	Unlimited (rate based)
Parameters	n, p	lambda only
Mean vs variance	Variance < mean	Mean = variance
Possible values	0 to n	0 to infinity

4. Normal Distribution

The most important continuous distribution - Gaussian , introduced by De Moivre and developed by Gauss, Laplace . Many natural variables (heights, marks, errors in measurement) are normal. By the central limit theorem sums / means of many independent r.v.s are approximately normal.

4.1 p.d.f.

$$f(x) = [\ 1 / (\sigma \sqrt{2\pi})\ ] \exp[\ -(x-\mu)^2 / (2\sigma^2)\ ],\ -\infty < x < \infty$$

X ~ N(mu, sigma^2). Parameters: mean mu and standard deviation sigma (sigma > 0).

4.2 Standard normal variate

$$Z = (X-\mu)/\sigma \sim N(0,1)\quad \phi(z) = (1/\sqrt{2\pi})\, e^{(-z^2/2)}$$

Tables give the area Phi(z) = P(Z <= z) or area from 0 to z. Use symmetry: P(Z <= -z) = 1 - P(Z <= z).

4.3 Properties of the normal curve

Properties

- Bell shaped, symmetric about x = mu; mean = median = mode = mu
- Total area under curve = 1; area on each side of mu = 0.5
- Points of inflection at x = mu +/- sigma; asymptotic to x-axis (never touches)
- Skewness = 0; Kurtosis = 3 (mesokurtic); mean deviation = 0.7979 sigma; quartile deviation = 0.6745 sigma
- M.G.F. = exp(mu t + sigma^2 t^2 / 2); central moments: mu_2 = sigma^2, odd moments = 0
- Sum of independent normals is normal: N(m1, s1^2) + N(m2, s2^2) = N(m1 + m2, s1^2 + s2^2)
- Maximum ordinate = 1/(sigma sqrt(2 pi)) at x = mu

Empirical rule (areas)

Interval	Area (probability)
mu +/- 1 sigma	0.6826 (68.26 %)
mu +/- 2 sigma	0.9544 (95.44 %)
mu +/- 3 sigma	0.9973 (99.73 %)
mu +/- 1.96 sigma	0.95 ; mu +/- 2.58 sigma: 0.99

Standard normal table values to remember

z	0.5	1.0	1.5	1.96	2.0	2.58	3.0
P(0<Z<z)	0.1915	0.3413	0.4332	0.4750	0.4772	0.4951	0.4987

Example 1

X ~ N(50, 10^2). P(45 < X < 62)? z1 = (45 - 50)/10 = -0.5, z2 = (62 - 50)/10 = 1.2. P = P(0<Z<0.5) + P(0<Z<1.2) = 0.1915 + 0.3849 = 0.5764 .

Example 2

Marks are normal with mean 60, sigma 10. P(X > 75): z = 1.5 -> 0.5 - 0.4332 = 0.0668 . If 500 students, about 33 score above 75. Top 10 % cut-off: z = 1.28 -> X = 60 + 12.8 = 72.8 .

Example 3 - finding mu and sigma

In a normal distribution 31 % of items are below 45 and 8 % are above 64. P(Z < -0.495) = 0.31 -> (45 - mu)/sigma = -0.495, P(Z > 1.405) = 0.08 -> (64 - mu)/sigma = 1.405. Subtract: 19/sigma = 1.9 -> sigma = 10 , mu = 45 + 4.95 = 49.95 (~50) .

Example 4 - bulbs

Life of bulbs ~ N(1200, 200^2) hours. P(X < 900) = P(Z < -1.5) = 0.0668. In 1000 bulbs, 67 expected to fail before 900 hrs. P(1000 < X < 1400) = 2 x 0.3413 = 0.6826 .

4.4 Normal approximation to binomial

For large n (n p > 5 and n q > 5): B(n, p) ~ N(n p, n p q). Apply continuity correction : P(X = K) ~ P(K - 1/2 < X < K + 1/2).

Example 5

A fair coin is tossed 100 times. P(48 to 55 heads)? mu = 50, sigma = 5. Continuity: 47.5 to 55.5 -> z = -0.5 to 1.1: 0.1915 + 0.3643 = 0.5558 .

Basis	Binomial	Poisson	Normal
Variable	Discrete	Discrete	Continuous
Shape	Skew unless p = 1/2	Skew, positive	Symmetric bell
Parameters	n, p	lambda	mu, sigma
Limit of	-	B(n,p), n large, p small	B(n,p), n large, p moderate

5. Other Useful Distributions (brief)

Uniform U(a, b)

f(x) = 1/(b - a) on a <= x <= b. Mean (a + b)/2, variance (b - a)^2/12.

Example: waiting time for a bus arriving uniformly every 10 min: P(wait > 7) = 3/10.

Exponential

f(x) = theta e^(-theta x), x > 0; mean 1/theta, variance 1/theta^2; memoryless. Time between Poisson events: if calls come at rate lambda per hour, the waiting time is exponential with theta = lambda.

Geometric

P(X = x) = q^(x-1) p, x = 1, 2, ... (trials until first success), mean 1/p, variance q/p^2.

Hypergeometric

Sampling without replacement: P(X = x) = C(M, x) C(N - M, n - x)/C(N, n). Becomes binomial for large N.

6. Derivations and Fitting

6.1 Binomial M.G.F. and moments

M(t) = Sum e^(tx) C(n,x) p^x q^(n-x) = Sum C(n,x)(p e^t)^x q^(n-x) = (q + p e^t)^n. Differentiate: M'(0) = n p (mean). M''(0) = n(n-1)p^2 + n p, so Var = n p - n p^2 = n p q.

6.2 Poisson mean and variance

E(X) = Sum x e^-l l^x/x! = l e^-l Sum l^(x-1)/(x-1)! = lambda. E[X(X-1)] = l^2 -> E(X^2) = l^2 + l -> Var = lambda.

6.3 Normal: area check

Put z = (x - mu)/sigma: Int f(x) dx = (1/sqrt(2 pi)) Int e^(-z^2/2) dz = (1/sqrt(2 pi)) sqrt(2 pi) = 1 (using Int e^(-a z^2) dz = sqrt(pi/a)).

6.4 Steps for a normal-distribution problem

Write X ~ N(mu, sigma^2) → Convert limits to z = (x − mu)/sigma → Sketch the curve and shade area

Read table values using symmetry → Add / subtract areas, multiply by N if frequency asked

Required	Use
P(Z < a), a > 0	0.5 + A(a)
P(Z > a), a > 0	0.5 − A(a)
P(−a < Z < b)	A(a) + A(b)
P(a < Z < b), 0 < a < b	A(b) − A(a)
P(Z < −a)	0.5 − A(a)

(A(z) = area from 0 to z in the table.)

6.5 Fitting a binomial distribution

Given observed data x with frequency f: (1) find mean = Sum f x / Sum f; (2) mean = n p → p = mean/n; (3) expected frequency = N × C(n,x) p^x q^(n−x).

Example - fit

5 coins tossed 320 times; mean number of heads found = 2.5 → p = 0.5. Expected frequencies: N P(x) = 320 × C(5,x)/32 = 10, 50, 100, 100, 50, 10 for x = 0 … 5.

6.6 Fitting a Poisson distribution

Mean lambda = Sum f x / Sum f. Expected frequency = N e^−l l^x/x!. Use recurrence: P(x + 1) = l/(x + 1) P(x).

x (deaths)	0	1	2	3	4
Observed f	109	65	22	3	1

N = 200; Sum f x = 0 + 65 + 44 + 9 + 4 = 122 → lambda = 0.61 . P(0) =

e^-0.61 = 0.5434 -> expected 108.7; P(1) = 0.3315 -> 66.3; P(2) = 0.1011 -> 20.2; P(3) = 0.0206 -> 4.1; P(4) = 0.0031 -> 0.6. Good fit with observed values.

6.7 More problems

(a) A machine makes 5 % defective parts. In a box of 20, P(exactly 1 defective) = C(20,1)(0.05)(0.95)^19 = 0.3774 (Binomial); Poisson approx with lambda = 1 gives e^-1 = 0.3679.

(b) Heights ~ N(165, 6^2) cm. P(height > 177) = P(Z > 2) = 0.0228 ; P(159 < X < 171) = 0.6826 .

(c) In 1 minute a switchboard gets calls Poisson(lambda = 2). P(more than 3 calls) = 1 - e^-2(1 + 2 + 2 + 1.333) = 1 - 0.8571 = 0.1429 .

(d) Weights of 1000 students ~ N(60, 8^2) Kg: number between 52 and 68 = 1000 x 0.6826 = 683 .

7. Practice Problems with Answers

Q	Problem	Answer
1	Binomial n = 8, p = 0.5: P(X = 5)	56/256 = 0.2188
2	Binomial n = 5, p = 1/3: mean and variance	5/3 ; 10/9
3	Poisson lambda = 4: P(X <= 1)	e^-4 (1 + 4) = 0.0916
4	Poisson lambda = 0.5: P(X = 0)	e^-0.5 = 0.6065
5	Z: P(-1.2 < Z < 0.8)	0.3849 + 0.2881 = 0.6730
6	N(100, 15^2): P(X > 130)	P(Z > 2) = 0.0228
7	Defect 2 %, sample 100: P(0 defects), Poisson	e^-2 = 0.1353
8	Binomial: 10 trials, p = 0.2, P(X >= 1)	1 - 0.8^10 = 0.8926

Quick revision - Unit IV

- Binomial: P(x) = C(n,x) p^x q^(n-x); mean np, variance npq
- Poisson: P(x) = e^-l l^x/x!; mean = variance = lambda; limit of binomial (n large, p small, np = lambda)
- Normal: bell shaped, mean = median = mode; Z = (X - mu)/sigma; 68-95-99.7 rule
- Normal approximation to binomial with continuity correction